

STC 122 - Practical 2 instructions

Instructions

File and Submission Requirements

- Answer the questions that follow and save all your code in a single R script.
- Name the script **Practical2.R**, where **Practical2** is the filename and **.R** is the file extension.
- Once you have completed your online submission, ensure the script is uploaded to clickUP using the correct file name, **Practical2.R**.
- Ensure all variables are named correctly. **Remember that variable names are case sensitive.**
- Ensure your code does not contain any syntax errors.
- Any code that is commented out will be treated as rough work.

Preparing for the Online Assessment

- The questions in this document are intended to prepare you for the online submission. Make sure you are comfortable with all questions here, as well as the content covered in the lecture notes, before attempting the online assessment.
- You will have **only ONE attempt** at the online submission, and it must be completed in **ONE sitting**. **You cannot save your progress and continue later. If you exit the attempt before submitting all your answers, you will be unable to resume and will forfeit the marks for the practical.**
- It is your responsibility to ensure you have a stable and sufficient internet connection before starting the online submission.

Academic Requirements

- You must include **detailed** explanations of your working as comments in your code, for **every** question in the assignment.
- You must also cite any resources you made use of while completing the assignment. Refer to the document `commenting_and_citing_in_your_code.pdf` on clickUP for guidance.
- You must complete the Declaration of Generative AI Use form after finishing your online submission. **Failure to submit this form will result in a mark of zero for the practical.**

Deadline

- The deadline for this practical is **Friday, 11 September 2026 at 23:59**.

Questions

- This assignment is based on the online Racer game from the Statistics Department at Grinnell College, Iowa, USA. It is not necessary to play the game in order to do this assignment, but see **Stat2games** to access the game and for more details regarding the game.
- Where applicable, answer the questions below by typing the appropriate **code** in the R script template provided on clickUP. Some questions are theoretical and no coding is needed to answer those questions.
- For this assignment you should use the dataset in the file `RacingGameData.csv` that is available on clickUP. The dataset contains the results of multiple races completed in the above-mentioned online racing game. The variables in the dataset are defined as follows:
 - Track: the track on which the race occurred.
 - Engine: the engine of the car used to complete the race.
 - FinishTime: the time, in seconds, to complete the race.

Question 1

Consider the continuous variable `FinishTime` for all races in the dataset.

- a) Calculate the sample mean for the finish times. Assign your answer to a variable called `Q1a`
- b) Calculate the sample standard deviation for the finish times. Assign your answer to a variable called `Q1b`
- c) Perform a Shapiro-Wilk test to determine if the finish times are normally distributed. Assign the resulting p -value to a variable called `Q1c`
- d) Based on the p -value calculated in Question 1(c), provide a decision for the normality test at a 5% level of significance. Assign the correct option to a variable called `Q1d`

For example, if the correct option is option E, assign "E" to a variable called `Q1d`:

```
Q1d <- "E"
```

A. Do not reject the null hypothesis and conclude that the finish times are normally distributed.
B. Reject the null hypothesis and conclude that the finish times are normally distributed.
C. Do not reject the null hypothesis and conclude that the finish times are not normally distributed.
D. Reject the null hypothesis and conclude that the finish times are not normally distributed.

Question 2

Assume that the population of finish times follows a normal distribution with a mean of $\mu = 27$ seconds and a standard deviation of $\sigma = 4.5$ seconds.

- Calculate the probability that a randomly selected race is completed in less than 25 seconds. Assign your answer to a variable called Q2a
- Calculate the probability that a randomly selected race takes longer than 30 seconds to complete. Assign your answer to a variable called Q2b
- Calculate the 10th percentile of the finish times (i.e., the time below which the fastest 10% of races are completed). Assign your answer to a variable called Q2c
- Calculate the Z-score associated with a sample mean of $\bar{x} = 25$ seconds. Assign your answer to a variable called Q2d

Question 3

Assume the population mean finish time is $\mu = 27$ seconds with a standard deviation of $\sigma = 4.5$ seconds. A random sample of $n = 40$ races is selected. Let $\bar{x}$ be the average finish time of these 40 races.

- Calculate the standard error of the sample mean ($\sigma_{\bar{x}}$). Assign your answer to a variable called Q3a
- Calculate the probability that the average finish time of these 40 races is less than 25 seconds. Assign your answer to a variable called Q3b

Question 4

We want to estimate the true population mean finish time for all races using bootstrapping on the provided dataset.

- Use bootstrapping to obtain simulated sample means for FinishTime. Use a seed value of 30 and 1000 bootstrap samples. Calculate the mean of these bootstrapped sample means and assign your answer to a variable called Q4a

- b) Calculate the lower bound of the 95% confidence interval for the mean finish time using the bootstrap percentile method. Assign your answer to a variable called Q4b
- c) Calculate the upper bound of the 95% confidence interval for the mean finish time using the bootstrap percentile method. Assign your answer to a variable called Q4c

HINT: Use the quantile function.

Question 5

For Question 5, consider only the races with the Bayes engine and the Nightingale engine.

Let μ_1 be the true mean finish time for races completed with the Bayes engine, and let μ_2 be the true mean finish time for races completed with the Nightingale engine. Perform a hypothesis test through randomisation to investigate whether there is a difference in the mean finish time between the two engines. Use a seed value of 35 and 1000 permutations.

- a) Select the option with the correct alternative hypothesis. Assign the correct option to a variable called Q5a

A.	$H_A : \mu_1 - \mu_2 > 0$
B.	$H_A : \bar{x}_1 - \bar{x}_2 \neq 0$
C.	$H_A : \mu_1 - \mu_2 \neq 0$
D.	$H_A : \mu_1 - \mu_2 < 0$

- b) Calculate the observed difference in sample means ($\bar{x}_1 - \bar{x}_2$). Assign your answer to a variable called Q5b
- c) Calculate the p -value based on your simulated null distribution. Assign your answer to a variable called Q5c
- d) Based on the p -value calculated in Question 5(c), provide a decision for the test. Assign the correct option to a variable called Q5d

A.	Do not reject the null hypothesis at a 5% level of significance and conclude that there is no significant difference in the mean finish times between the two engines.
B.	Reject the null hypothesis at a 5% level of significance and conclude that there is a significant difference in the mean finish times between the two engines.
C.	Do not reject the null hypothesis at a 5% level of significance and conclude that there is a significant difference in the mean finish times between the two engines.
D.	Reject the null hypothesis at a 5% level of significance and conclude that there is no significant difference in the mean finish times between the two engines.